

Six years, ~73,000 trades, two universes — and an honest read on what it proves

The tuned strategist is profitable and statistically well-separated from chance across both large-cap and small-cap universes. The binding constraint on confidence isn't sample size — it's that option premiums are modeled, not quoted.

Window Oct 2020 – Jul 2026 **Universe** 500 S&P + 127 small-cap **Tenor** 35 trading days, rolled every 10
Report 10–11 Jul 2026 (incl. overnight validation suite)

Headline result

● 10-gate validation: 7 pass · 1 marginal · 1 flag · 1 pending

S&P 500 — RETURN ON RISK

+21.5%

52.6% win rate over 60,954 trades. Every regime bucket positive, including a bear regime that now returns **+17.2%**.

SMALL-CAP — RETURN ON RISK

+7.9%

61.4% win rate over 11,985 trades. Weaker in absolute terms, but well-calibrated and positive in every regime.

Performance by structure

Return on risk = mean P&L as a fraction of the capital at risk per trade — the unit that's comparable across a \$60 stock and a \$600 one. Win rate and profitability diverge by design: directional debit spreads win under half the time but pay more on wins than they lose.

S&P 500 STRUCTURE	TRADES	WIN %	PRED POP	RETURN / RISK
Long Straddle	25,558	50.9%	42.7%	+24.4%
Call Debit Spread	29,881	49.2%	42.1%	+21.2%
Bull Put Spread	1,695	81.8%	69.4%	+16.7%
Iron Condor	1,732	77.9%	63.2%	+16.6%
Bear Call Spread	62	91.9%	81.8%	+11.8%
Cash-Secured Put	1,712	84.6%	72.5%	+2.2%
Put Debit Spread	314	31.5%	44.5%	-27.0%
All structures	60,954	52.6%	44.6%	+21.5%

Put Debit Spread (−27%) is the one known residual: 314 trades, 0.5% of volume — negligible drag, deliberately not chased to avoid over-fitting.

SMALL-CAP STRUCTURE	TRADES	WIN %	PRED POP	RETURN / RISK
Long Straddle	2,827	44.1%	42.4%	+17.9%
Iron Condor	793	74.7%	63.4%	+14.6%
Cash-Secured Put	3,226	74.2%	69.6%	+6.9%
Bull Put Spread	2,891	68.5%	66.5%	+3.5%
Bear Call Spread	372	83.1%	81.4%	+2.4%
Put Debit Spread	1,876	44.6%	46.2%	-0.4%
All structures	11,985	61.4%	58.7%	+7.9%

Small-cap preset favours selling premium (CSP, credit spreads) over buying direction — richer premium, no reliance on a steady upward grind that small-caps don't have.

What the tuning bought (S&P)

Three rounds of changes, each held to the same 6-year backtest. The bear-regime line is the standout — it flipped from a persistent loss to the second-strongest regime.

Regime gate only		+14.4%
+ 3 core levers		+19.8%
+ bear gate & size preset		+21.5%

Bear regime, before → after: -7.6% → +17.2%. High-conviction bearish trades turned out to be *anti-predictive* in a bear tape — the strongest downtrends snap back hardest — so they now route to neutral structures, which earned +11–27% in exactly those windows.

Is the edge real, or sampling noise?

Win rates below, with 95% intervals computed on an **overlap-adjusted** sample. Because a 35-day trade is re-entered every 10 days, consecutive trades on a name overlap $\sim 3.5\times$, so the effective independent count is roughly $N \div 3.5$ — a deliberately conservative haircut. The σ column is standard deviations above a coin-flip.

BUCKET	TRADES	EFF. N	WIN %	95% INTERVAL	Σ VS 50%
S&P — all	60,954	17,415	52.6%	51.9 – 53.3%	+6.9
S&P — bear regime	4,020	1,149	56.0%	53.1 – 58.8%	+4.1
Small-cap — all	11,985	3,424	61.4%	59.8 – 63.0%	+13.7
Iron Condor (S&P)	1,732	495	77.9%	74.0 – 81.3%	+15.0
Bull Put (S&P)	1,695	484	81.8%	78.1 – 85.0%	+18.1
Call Debit (S&P)	29,881	8,537	49.2%	48.1 – 50.3%	-1.5

Call Debit's win rate straddles 50% — expected, and not a weakness: its edge lives in payoff asymmetry (wins bigger than losses), which is why it still returns +21% on risk. Win rate is the wrong lens for directional structures.

Verdict on significance: every aggregate and every premium-selling structure clears 4σ even after the overlap haircut — the win-rate edge is not sampling noise. What that does *not* settle is whether the edge survives real fills.

Profit factor & risk-adjusted return

Win rate ignores *magnitude*. These metrics don't. Computed on 17,923 per-trade returns from a representative 150-name S&P subset — whose mean return-on-risk of **+21.7%** reproduces the full-universe **+21.5%**, confirming it's not cherry-picked. The 95% intervals come from a **symbol-block bootstrap** (resampling whole names, 2,000 draws) — the honest way to price in the overlap correlation, stronger than the win-rate haircut above.

MEAN RETURN ON RISK

+21.7%

95% CI [+19.2%, +24.1%]

PROFIT FACTOR

1.62

95% CI [1.54, 1.70]

SHARPE · PER TRADE

0.20

95% CI [0.18, 0.22]

Reading these: a profit factor of **1.62** means gross winners out-earn gross losers by ~62% — comfortably above the 1.0 breakeven, with the whole interval clear of it. The bootstrap interval on return also sits entirely above zero, so it's now the *profitability*, not just the win rate, that's statistically separated from noise. Per-trade Sharpe annualizes to roughly **1.0** — respectable — but overlap makes that scaling optimistic, so treat it as a ceiling, not a promise.

STRUCTURE (S&P SUBSET)	TRADES	PROFIT FACTOR	SHARPE / TRADE	MEAN R
Cash-Secured Put	459	2.59	0.362	+2.1%
Long Straddle	7,661	1.98	0.247	+24.6%
Bull Put Spread	453	1.92	0.292	+14.0%
Iron Condor	450	1.77	0.264	+14.8%
Call Debit Spread	8,869	1.45	0.173	+21.0%

Cash-Secured Put shows the paradox cleanly: highest profit factor and Sharpe (small, reliable wins) but the lowest return on risk (huge collateral). Long Straddle is the standout — high magnitude *and* high consistency.

Validation suite: ten acceptance gates

A professional-grade gate list (walk-forward, baselines, plateaus, robustness, drawdown) run against the tuned strategy. Every gate that *can* be tested on synthetic data has been; what remains needs real quotes and time.

ACCEPTANCE GATE		RE
OOS expectancy positive	OOS +22.8% gross; ~+11.5% at 5% slip	
Lower confidence bound > 0	Bootstrap CI [+19.2%, +24.1%]	
Survives 2× costs	Breakeven at ~11% round-trip slip	
Parameter plateaus, not cliffs	All 6 neighbors +20.1–23.4%	
No single year / symbol dominates	Max year 31%, top symbol 1.9% — but Long Straddle 48.5% of	
OOS retains ≥ ~50%	110% (S&P), 87% (small-cap)	
Drawdown fits risk budget	MDD ≈ 2.2–2.8× total risk cap; 385 days underwater	
Beats the simple baseline	3× the alpha at 40% of the beta (see below)	
Not merely equity beta	+14.6pt alpha vs +7.1pt β×market	
Paper fills ≈ modeled fills	Chain recorder live (109 symbols daily); needs months	

Walk-forward: does the edge survive out-of-sample?

The levers were developed on 2020–2023 patterns. Frozen and applied to 2024–2026 — data no lever value ever saw — the strategy performs *better*, and each lever passes the harder test: it would have been adopted on in-sample data alone *and* still adds out-of-sample.

CONFIG	IS MEAN R	IS PF	OOS MEAN R	OOS PF	RETENTION
FINAL (S&P)	+20.7%	1.57	+22.8%	1.67	110%
FINAL (small-cap)	+9.4%	1.40	+8.2%	1.36	87%
without EV gate	+17.4%	1.48	+20.3%	1.59	lever +2.5pt OOS
without drift	+14.5%	1.48	+15.9%	1.51	lever +6.9pt OOS
without bear gates	+14.5%	1.36	+20.8%	1.60	lever +2.0pt OOS

Baseline: is the tuning worth anything?

Two mechanical comparators on the same chains, same dates, no signals. The raw return of "always buy an ATM call debit" nearly matches the tuned strategy — but decomposition shows it's ~80% levered market exposure. The tuned strategist earns **3× the alpha at 40% of the beta**; the mechanical premium-seller loses outright.

	MEAN R	PF	SHARPE/TRADE	ALPHA	BETA
Tuned strategist	+21.7%	1.62	0.200	+14.6%	3.3
Always-buy call debit	+21.0%	1.41	0.160	+4.5%	7.8
Always-sell iron condor	-6.7%	0.81	—	—	—

Costs, plateaus, robustness, drawdown

- **Cost stress:** at 2.5% round-trip slippage PF 1.42 / +16.0%; at 5% PF 1.29 / +11.5%; breakeven ≈ 11%; at 15% the strategy loses. The edge survives realistic costs but not sloppy execution — liquid, tight-spread underlyings only.
- **Parameter plateau:** every neighbor of the tuned point (drift 4/12, delta .25/.35, DTE 25/45) earns +20.1% to +23.4% — a smooth plateau, no curve-fit cliff. DTE 45 tested mildly better; left alone on the plateau principle.

- **Robustness:** no year supplies more than 31% of P&L, top symbol 1.9%, top five 9.0%. Honest flags: Long Straddle contributes 48.5% of P&L and the top 5% of trades supply 55% — a long-vol, tail-driven profit engine that deserves its own ask-side-fill validation before real money.
- **Drawdown (scale-invariant):** max drawdown $\approx 2.2\text{--}2.8\times$ the standing total-risk cap; longest underwater stretch **385 days at any size**; 26% of months negative. At a 5% total-risk cap: -13.5% MDD, Calmar ~ 3.4 , $\sim +22\%$ CAGR after realistic slippage. Marks at expiry only, so true intra-trade MDD is deeper.
- **Honest portfolio Sharpe:** 0.57 on non-overlapping cohorts (the 1.47 overlapping figure is the optimistic bound).

Confidence, along four axes

A single number would flatter the result. Confidence is high on some dimensions and genuinely limited on others — and the limited ones are structural, not fixable with more data.

Statistical significance

HIGH 

60,954 S&P trades; win rate holds at $+6.9\sigma$ after a conservative overlap haircut, and a symbol-block bootstrap puts the return-on-risk interval at $[+19.2\%, +24.1\%]$ — profitability itself, not just win rate, is clear of zero.

Over-fitting risk

VERY LOW 

Now evidenced, not just argued: 110% out-of-sample retention, every lever valid on in-sample data alone, and a smooth parameter plateau (all neighbors $+20\text{--}23\%$). Curve-fit strategies do not do this.

Regime coverage

MODERATE 

~93% of trades fall in bull regimes. The bull result is deeply sampled; the bear result (~4k trades) is positive and significant but rests on one 2022-era stretch — thinner evidence.

Transfer to live P&L

LIMITED, NOW BOUNDED 

Premiums are still modeled, but the cost stress bounds the exposure: the edge survives 5% round-trip slippage ($+11.5\%$) and dies at $\sim 11\%$. A daily recorder now archives real chains (109 symbols, sector-stratified) — the path to closing this axis with paper-fill comparison.

What this report does *not* prove

The single biggest caveat: **no historical option chains exist for this account**, so entry premiums are priced by Black-Scholes off trailing realized volatility. Any edge that lives in **real implied vol vs realized** — a large part of live options P&L — is assumed away here.

- **No transaction costs.** Commissions, bid-ask slippage, and early-assignment risk are unmodeled; real net returns sit below these figures.
- **Hold-to-expiry marks.** The profit-taking and stop rules aren't simulated — conservative for premium sellers, generous for debit buyers.
- **Bull-heavy sample.** The window is dominated by a rising market; sustained-bear behaviour is thinly tested.
- **In-sample.** Parameters were chosen on this same 6-year window — low over-fit risk given how few there are, but not a walk-forward out-of-sample test.

Bottom line

The selection logic is validated **as far as synthetic data can take it**: out-of-sample (110% retention), against baselines (3× the alpha at 40% of the beta), across parameter neighborhoods (plateau, no cliff), net of modeled costs (survives 5% slippage), and decomposed from market beta. Seven of ten acceptance gates pass; none fail. The honest expectation is a **modest, real edge** — portfolio Sharpe ~0.6, a quarter of months negative, year-long underwater stretches — not a money printer. What remains is exactly what no simulation can supply: months of real quotes (now recording daily, 109 symbols), a paper-fill comparison, and a pilot sized so the worst case costs ~0.2% of capital. Nothing left to compute; from here, only real quotes and time raise confidence.

Generated from runs saved at `data_store/reports/backtest_FINAL_sp500_2026-07-10.txt` and ...
`_smallcap_2026-07-10.txt`. Return-on-risk and win rates as reported by the synthetic backtest. Win-rate intervals are Wilson score intervals on the overlap-adjusted effective sample (haircut 3.5×); profit factor, Sharpe, and

return intervals are symbol-block bootstraps (2,000 resamples of 17,923 per-trade returns across 150 names).
Methodology caveats above bound all figures.